

3.1 Probabilistic Inference in Structured Distributions

Consider an environment composed of N variables, with a corresponding distribution $p(x_1, \dots, x_N)$. Writing $\mathcal{E}$ as the set of evidential variables and using $\text{evidence} = \{x_e = x_e, e \in \mathcal{E}\}$ to denote all available evidence, then inference and reasoning can be carried out automatically by the ‘brute force’ method¹

$$p(x_i = x_i | \text{evidence}) = \frac{\sum_{x_{\{\mathcal{E}, \setminus i\}}} p(\{x_e = x_e, e \in \mathcal{E}\}, x_i = x_i, x_{\{\mathcal{E}, \setminus i\}})}{\sum_{x_{\setminus \mathcal{E}}} p(\{x_e = x_e, e \in \mathcal{E}\}, x_{\setminus \mathcal{E}})} \quad (3.1.1)$$

If all variables are binary (take two states), these summations require $O(2^{N-|\mathcal{E}|})$ operations. Such exponential computation is impractical and techniques that reduce this burden by exploiting any structure in the joint probability table are the topic of our discussions on efficient inference.

Naively specifying all the entries of a table $p(x_1, \dots, x_N)$ over binary variables x_i takes $O(2^N)$ space. We will need to deal with large numbers of variables in machine learning and related application areas, with distributions on potentially hundreds if not millions of variables. The only way to deal with such large distributions is to constrain the nature of the variable interactions in some manner, both to render specification and ultimately inference in such systems tractable. The key idea is to specify which variables are independent of others, leading to a structured factorisation of the joint probability distribution. Belief Networks are a convenient framework for representing such factorisations into local conditional distributions. We will discuss Belief Networks more formally in section(3.3), first discussing their natural graphical representations of distributions.

Definition 18 (Belief Network). A Belief Network is a distribution of the form

$$p(x_1, \dots, x_D) = \prod_{i=1}^D p(x_i | \text{pa}(x_i)) \quad (3.1.2)$$

where $\text{pa}(x_i)$ represent the *parental* variables of variable x_i . Written as a Directed Graph, with an arrow pointing from a parent variable to child variable, a Belief Network is a Directed Acyclic Graph (DAG), with the i^{th} vertex in the graph corresponding to the factor $p(x_i | \text{pa}(x_i))$.

¹The extension to continuous variables is straightforward, replacing summation with integration over pdfs; we defer treatment of this to later chapters, since our aim is to here outline more the intuitions without needing to deal with integration of high dimensional distributions.

3.2 Graphically Representing Distributions

Belief Networks (also called Bayes' Networks or Bayesian Belief Networks) are a way to depict the independence assumptions made in a distribution [148, 168]. Their application domain is widespread, ranging from troubleshooting[50] and expert reasoning under uncertainty to machine learning. Before we more formally define a BN, an example will help motivate the development².

3.2.1 Constructing a simple Belief network : wet grass

One morning Tracey leaves her house and realises that her grass is wet. Is it due to overnight rain or did she forget to turn off the sprinkler last night? Next she notices that the grass of her neighbour, Jack, is also wet. This *explains away* to some extent the possibility that her sprinkler was left on, and she concludes therefore that it has probably been raining.

Making a model

We can model the above situation using probability by following a general modelling approach. First we define the variables we wish to include in our model. In the above situation, the natural variables are

$R \in \{0, 1\}$ ($R = 1$ means that it has been raining, and 0 otherwise).

$S \in \{0, 1\}$ ($S = 1$ means that Tracey has forgotten to turn off the sprinkler, and 0 otherwise).

$J \in \{0, 1\}$ ($J = 1$ means that Jack's grass is wet, and 0 otherwise).

$T \in \{0, 1\}$ ($T = 1$ means that Tracey's Grass is wet, and 0 otherwise).

A model of Tracey's world then corresponds to a probability distribution on the joint set of the variables of interest $p(T, J, R, S)$ (the order of the variables is irrelevant).

Since each of the variables in this example can take one of two states, it would appear that we naively have to specify the values for each of the $2^4 = 16$ states, *e.g.* $p(T = 1, J = 0, R = 1, S = 1) = 0.7$ *etc.* However, since there are normalisation conditions for probabilities, we do not need to specify all the state probabilities. To see how many states need to be specified, consider the following decomposition. Without loss of generality (wlog) and repeatedly using Bayes' rule, we may write

$$p(T, J, R, S) = p(T|J, R, S)p(J, R, S) \quad (3.2.1)$$

$$= p(T|J, R, S)p(J|R, S)p(R, S) \quad (3.2.2)$$

$$= p(T|J, R, S)p(J|R, S)p(R|S)p(S) \quad (3.2.3)$$

That is, we may write the joint distribution as a product of conditional distributions. The first term $p(T|J, R, S)$ requires us to specify $2^3 = 8$ values – we need $p(T = 1|J, R, S)$ for the 8 joint states of J, R, S . The other value $p(T = 0|J, R, S)$ is given by normalisation : $p(T = 0|J, R, S) = 1 - p(T = 1|J, R, S)$. Similarly, we need $4 + 2 + 1$ values for the other factors, making a total of 15 values in all. In general, for a distribution on n binary variables, we need to specify $2^n - 1$ values in the range $[0, 1]$. The important point here is that the number of values that need to be specified in general scales exponentially with the number of variables in the model – this is impractical in general and motivates simplifications.

Conditional independence

The modeler often knows constraints on the system. For example, in the scenario above, we may assume that Tracey's grass is wet depends only directly on whether or not it has been raining and whether or not her sprinkler was on. That is, we make a *conditional independence* assumption

$$p(T|J, R, S) = p(T|R, S) \quad (3.2.4)$$

²The scenario is adapted from [219].

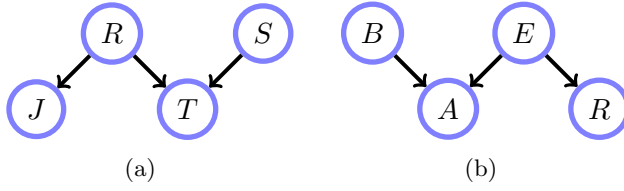


Figure 3.1: **(a)**: Belief network structure for the ‘wet grass’ example. Each node in the graph represents a variable in the joint distribution, and the variables which feed in (the parents) to another variable represent which variables are to the right of the conditioning bar. **(b)**: BN for the Burglar model.

Similarly, since whether or not Jack’s grass is wet is influenced only directly by whether or not it has been raining, we write

$$p(J|R, S) = p(J|R) \quad (3.2.5)$$

and since the rain is not directly influenced by the sprinkler,

$$p(R|S) = p(R) \quad (3.2.6)$$

which means that our model now becomes :

$$p(T, J, R, S) = p(T|R, S)p(J|R)p(R)p(S) \quad (3.2.7)$$

We can represent these conditional independencies graphically, as in fig(3.1a). This reduces the number of values that we need to specify to $4 + 2 + 1 + 1 = 8$, a saving over the previous 15 values in the case where no conditional independencies had been assumed.

To complete the model, we need to numerically specify the values of each conditional probability table (CPT). Let the prior probabilities for R and S be $p(R = 1) = 0.2$ and $p(S = 1) = 0.1$. We set the remaining probabilities to $p(J = 1|R = 1) = 1$, $p(J = 1|R = 0) = 0.2$ (sometimes Jack’s grass is wet due to unknown effects, other than rain), $p(T = 1|R = 1, S) = 1$, $p(T = 1|R = 0, S = 1) = 0.9$ (there’s a small chance that even though the sprinkler was left on, it didn’t wet the grass noticeably), $p(T = 1|R = 0, S = 0) = 0$.

Inference

Now that we’ve made a model of an environment, we can perform inference. Let’s calculate the probability that the sprinkler was on overnight, given that Tracey’s grass is wet: $p(S = 1|T = 1)$.

To do this, we use Bayes’ rule:

$$p(S = 1|T = 1) = \frac{p(S = 1, T = 1)}{p(T = 1)} = \frac{\sum_{J,R} p(T = 1, J, R, S = 1)}{\sum_{J,R,S} p(T = 1, J, R, S)} \quad (3.2.8)$$

$$= \frac{\sum_{J,R} p(J|R)p(T = 1|R, S = 1)p(R)p(S = 1)}{\sum_{J,R,S} p(J|R)p(T = 1|R, S)p(R)p(S)} \quad (3.2.9)$$

$$= \frac{\sum_R p(T = 1|R, S = 1)p(R)p(S = 1)}{\sum_{R,S} p(T = 1|R, S)p(R)p(S)} \quad (3.2.10)$$

$$= \frac{0.9 \times 0.8 \times 0.1 + 1 \times 0.2 \times 0.1}{0.9 \times 0.8 \times 0.1 + 1 \times 0.2 \times 0.1 + 0 \times 0.8 \times 0.9 + 1 \times 0.2 \times 0.9} = 0.3382 \quad (3.2.11)$$

so the belief that the sprinkler is on increases above the prior probability 0.1, due to the fact that the grass is wet.

Let us now calculate the probability that Tracey’s sprinkler was on overnight, given that her grass is wet and that Jack’s grass is also wet, $p(S = 1|T = 1, J = 1)$. We use Bayes’ rule again:

$$p(S = 1|T = 1, J = 1) = \frac{p(S = 1, T = 1, J = 1)}{p(T = 1, J = 1)} \quad (3.2.12)$$

$$= \frac{\sum_R p(T = 1, J = 1, R, S = 1)}{\sum_{R,S} p(T = 1, J = 1, R, S)} \quad (3.2.13)$$

$$= \frac{\sum_R p(J = 1|R)p(T = 1|R, S = 1)p(R)p(S = 1)}{\sum_{R,S} p(J = 1|R)p(T = 1|R, S)p(R)p(S)} \quad (3.2.14)$$

$$= \frac{0.0344}{0.2144} = 0.1604 \quad (3.2.15)$$

The probability that the sprinkler is on, given the extra evidence that Jack's grass is wet, is *lower* than the probability that the grass is wet given only that Tracey's grass is wet. That is, that the grass is wet due to the sprinkler is (partly) explained away by the fact that Jack's grass is also wet – this increases the chance that the rain has played a role in making Tracey's grass wet.

Naturally, we don't wish to carry out such inference calculations by hand all the time. General purpose algorithms exist for this, such as the Junction Tree Algorithm, and we shall introduce these in later chapters.

Example 10 (Was it the Burglar?). Here's another example using binary variables, adapted from [219]. Sally comes home to find that the burglar alarm is sounding ($A = 1$). Has she been burgled ($B = 1$), or was the alarm triggered by an earthquake ($E = 1$)? She turns the car radio on for news of earthquakes and finds that the radio broadcasts an earthquake alert ($R = 1$).

Using Bayes' rule, we can write, without loss of generality,

$$p(B, E, A, R) = p(A|B, E, R)p(B, E, R) \quad (3.2.16)$$

We can repeat this for $p(B, E, R)$, and continue

$$p(B, E, A, R) = p(A|B, E, R)p(R|B, E)p(E|B)p(B) \quad (3.2.17)$$

However, the alarm is surely not directly influenced by any report on the Radio – that is, $p(A|B, E, R) = p(A|B, E)$. Similarly, we can make other conditional independence assumptions such that

$$p(B, E, A, R) = p(A|B, E)p(R|E)p(E)p(B) \quad (3.2.18)$$

Specifying conditional probability tables

Alarm = 1	Burglar	Earthquake
0.9999	1	1
0.99	1	0
0.99	0	1
0.0001	0	0

Radio = 1	Earthquake
1	1
0	0

The remaining tables are $p(B = 1) = 0.01$ and $p(E = 1) = 0.000001$. The tables and graphical structure fully specify the distribution. Now consider what happens as we observe evidence.

Initial Evidence: The Alarm is sounding

$$p(B = 1|A = 1) = \frac{\sum_{E,R} p(B = 1, E, A = 1, R)}{\sum_{B,E,R} p(B, E, A = 1, R)} \quad (3.2.19)$$

$$= \frac{\sum_{E,R} p(A = 1|B = 1, E)p(B = 1)p(E)p(R|E)}{\sum_{B,E,R} p(A = 1|B, E)p(B)p(E)p(R|E)} \approx 0.99 \quad (3.2.20)$$

Additional Evidence: The Radio broadcasts an Earthquake warning: A similar calculation gives $p(B = 1|A = 1, R = 1) \approx 0.01$. Thus, initially, because the Alarm sounds, Sally thinks that she's been burgled. However, this probability drops dramatically when she hears that there has been an Earthquake. That is, the Earthquake 'explains away' to an extent the fact that the Alarm is ringing. See `demoBurglar.m`.

3.2.2 Uncertain evidence

In soft or *uncertain evidence*, the variable is in more than one state, with the strength of our belief about each state being given by probabilities. For example, if x has the states $\text{dom}(x) = \{\text{red}, \text{blue}, \text{green}\}$ the vector $(0.6, 0.1, 0.3)$ represents the probabilities of the respective states. In contrast, for *hard evidence* we are certain that a variable is in a particular state. In this case, all the probability mass is in one of the vector components, for example $(0, 0, 1)$.

Performing inference with soft-evidence is straightforward and can be achieved using Bayes' rule. Writing the soft evidence as $\tilde{y}$, we have

$$p(x|\tilde{y}) = \sum_y p(x|y)p(y|\tilde{y}) \quad (3.2.21)$$

where $p(y = i|\tilde{y})$ represents the probability that y is in state i under the soft-evidence. This is a generalisation of hard-evidence in which the vector $p(y|\tilde{y})$ has all zero component values, except for all but a single component.

Note that the soft evidence $p(y = i|\tilde{y})$ does not correspond to the marginal $p(y = i)$ in the original joint distribution $p(x, y)$. A procedure to form a joint distribution, known as *Jeffrey's rule* is to begin with an original distribution $p_1(x, y)$, from which we can define

$$p_1(x|y) = \frac{p_1(x, y)}{\sum_x p_1(x, y)} \quad (3.2.22)$$

Using the soft evidence $p(y|\tilde{y})$ we then define a new joint distribution

$$p_2(x, y|\tilde{y}) = p_1(x|y)p(y|\tilde{y}) \quad (3.2.23)$$

In the BN we use a dashed circle to represent that a variable is in a soft-evidence state.

Example 11 (soft-evidence). Revisiting the earthquake scenario, `example(10)`, imagine that we think we hear the burglar alarm sounding, but are not sure, specifically we are only 70% sure we heard the alarm. For this binary variable case we represent this soft-evidence for the states $(1, 0)$ as $\tilde{A} = (0.7, 0.3)$. What is the probability of a burglary under this soft-evidence?

$$p(B = 1|\tilde{A}) = \sum_A p(B = 1|A)p(A|\tilde{A}) = p(B = 1|A = 1) \times 0.7 + p(B = 1|A = 0) \times 0.3 \quad (3.2.24)$$

The probabilities $p(B = 1|A = 1) \approx 0.99$ and $p(B = 1|A = 0) \approx 0.0001$ are calculated using Bayes' rule as before to give

$$p(B = 1|\tilde{A}) \approx 0.6930 \quad (3.2.25)$$

Uncertain evidence versus unreliable modelling

An entertaining example of uncertain evidence is given by Pearl[219]:

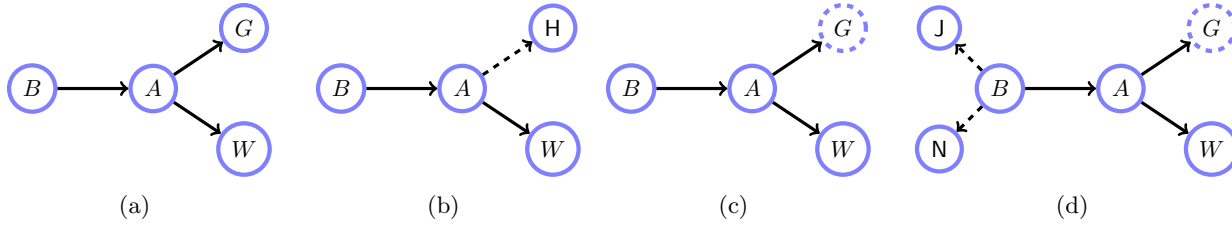


Figure 3.2: **(a)**: Mr Holmes' burglary worries as given in [219]: (B)urglar, (A)larm, (W)atson, Mrs (G)ibbon. **(b)**: Virtual Evidence can be represented by a dashed line. **(c)**: Modified problem. Mrs Gibbon is not drinking but somewhat deaf; we represent such uncertain (soft-evidence) by a circle. **(d)**: Holmes gets additional information from his neighbour Mrs (N)osy and informant Dodgy (J)oe.

Mr Holmes receives a telephone call from his neighbour Dr Watson, who states that he hears the sound of a burglar alarm from the direction of Mr Holmes' house. While preparing to rush home, Mr Holmes recalls that Dr Watson is known to be a tasteless practical joker, and he decides to first call another neighbour, Mrs Gibbon who, despite occasional drinking problems, is far more reliable.

When Mr Holmes calls Mrs Gibbon, he soon realises that she is somewhat tipsy. Instead of answering his question directly, she goes on and on about her latest back operation and about how terribly noisy and crime-ridden the neighbourhood has become. When he finally hangs up, all Mr Holmes can glean from the conversation is that there is probably an 80% chance that Mrs Gibbon did hear an alarm sound from her window.

A BN for this scenario is depicted in fig(3.2a) which deals with four binary variables: House is (B)urgled, (A)larm has sounded, (W)atson hears alarm, and Mrs (G)ibbon hears alarm³:

$$p(B, A, G, W) = p(A|B)p(B)p(W|A)p(G|A) \quad (3.2.26)$$

Holmes is interested in the likelihood that his house has been burgled. Naively, Holmes' might calculate⁴

$$p(B = \text{tr} | W = \text{tr}, G = \text{tr}) \quad (3.2.27)$$

However, after finding out about Mrs Gibbon's state, Mr Holmes no longer finds the above model reliable. He wants to ignore the effect that Mrs Gibbon's evidence has on the inference, and replace it with his own belief as to what Mrs Gibbon observed. Mr Holmes can achieve this by replacing the term $p(G = \text{tr} | A)$ by a so-called *virtual evidence* term

$$p(G = \text{tr} | A) \rightarrow p(H|A), \quad \text{where } p(H|A) = \begin{cases} 0.8 & A = \text{tr} \\ 0.2 & A = \text{fa} \end{cases} \quad (3.2.28)$$

Here the state H is arbitrary and fixed. This is used to modify the joint distribution to

$$p(B, A, H, W) = p(A|B)p(B)p(W|A)p(H|A), \quad (3.2.29)$$

see fig(3.2b). When we then compute $p(B = \text{tr} | W = \text{tr}, H)$ the effect of Mr Holmes' judgement will count for a factor of 4 times more in favour of the Alarm sounding than not. The values of the table entries are irrelevant up to normalisation since any constants can be absorbed into the proportionality constant. Note also that $p(H|A)$ is not a distribution in A , and hence no normalisation is required. This form of evidence is also called *likelihood evidence*.

A twist on Pearl's scenario is that Mrs Gibbon has not been drinking. However, she is a little deaf and cannot be sure herself that she heard the alarm. She is 80% sure she heard it. In this case, Holmes would

³One might be tempted to include an additional (T)ipsy variable as a parent of G . This would then require us to specify the joint distribution $p(G|T, A)$ for the 4 parental joint states of T and A . Here we assume that we do not have access to such information.

⁴The notation tr is equivalent to 1 and fa to 0 from example(10).

trust the model – however, the observation itself is now uncertain, fig(3.2c). This can be dealt with using the soft evidence technique. From Jeffrey’s rule, one uses the original model equation (3.2.26) to compute

$$p(B = \text{tr}|W = \text{tr}, G) = \frac{p(B = \text{tr}, W = \text{tr}, G)}{p(W = \text{tr}, G)} = \frac{\sum_A p(G|A)p(W = \text{tr}|A)p(A|B = \text{tr})}{\sum_{B,A} p(G|A)p(W = \text{tr}|A)p(A|B)} \quad (3.2.30)$$

and then uses the soft-evidence

$$p(G|\tilde{G}) = \begin{cases} 0.8 & G = \text{tr} \\ 0.2 & G = \text{fa} \end{cases} \quad (3.2.31)$$

to compute

$$p(B = \text{tr}|W = \text{tr}, \tilde{G}) = p(B = \text{tr}|W = \text{tr}, G = \text{tr})p(G = \text{tr}|\tilde{G}) + p(B = \text{tr}|W = \text{tr}, G = \text{fa})p(G = \text{fa}|\tilde{G}) \quad (3.2.32)$$

The reader may show that an alternative way to represent an uncertain observation such as Mrs Gibbon being non-tipsy but hard-of-hearing above is to use a virtual evidence child from G .

Uncertain evidence within an unreliable model

To highlight uncertain evidence in an unreliable model we introduce two additional characters. Mrs Nosy lives next door to Mr Holmes and is completely deaf, but nevertheless an incorrigible curtain-peeker who seems to notice most things. Unfortunately, she’s also rather prone to imagining things. Based on his conversation with her, Mr Holmes counts her story as 3 times in favour of there not being a burglary to there being a burglary, and therefore uses a virtual evidence term

$$p(\text{Nosy}|B) = \begin{cases} 1 & B = \text{tr} \\ 3 & B = \text{fa} \end{cases} \quad (3.2.33)$$

Mr Holmes also telephones Dodgy Joe, his contact in the criminal underworld to see if he’s heard of any planned burglary on Mr Holmes’ home. He summarises this information using a virtual evidence term

$$p(\text{Joe}|B) = \begin{cases} 1 & B = \text{tr} \\ 5 & B = \text{fa} \end{cases} \quad (3.2.34)$$

When all this information is combined : Mrs Gibbon is not tipsy but somewhat hard of hearing, Mrs Nosy, and Dodgy Joe, we first deal with the unreliable model

$$p(B, A, W = \text{tr}, G, \text{Nosy}, \text{Joe}) \propto p(B)p(\text{Nosy}|B)p(\text{Joe}|B)p(A|B)p(W = \text{tr}|A)p(G|A) \quad (3.2.35)$$

from which we can compute

$$p(B = \text{tr}|W = \text{tr}, G, \text{Nosy}, \text{Joe}) \quad (3.2.36)$$

Finally we perform inference with the soft-evidence

$$p(B = \text{tr}|W = \text{tr}, \tilde{G}, \text{Nosy}, \text{Joe}) = \sum_G p(B = \text{tr}|W = \text{tr}, G, \text{Nosy}, \text{Joe})p(G|\tilde{G}) \quad (3.2.37)$$

An important consideration above is that the virtual evidence does not replace the prior $p(B)$ with another prior distribution – rather the virtual evidence terms modify the prior through the inclusion of extra factors. The usual assumption is that each virtual evidence acts independently, although one can consider dependent scenarios if required.



Figure 3.3: Two BNs for a 4 variable distribution. Both graphs (a) and (b) represent the *same* distribution $p(x_1, x_2, x_3, x_4)$. Strictly speaking they represent the same (lack of) independence assumptions – the graphs say nothing about the content of the CPTs. The extension of this ‘cascade’ to many variables is clear and always results in a Directed Acyclic Graph.

3.3 Belief Networks

In the Wet Grass and Burglar examples, we had a choice as to how we recursively used Bayes’ rule. In a general 4 variable case we could choose the factorisation,

$$p(x_1, x_2, x_3, x_4) = p(x_1|x_2, x_3, x_4)p(x_2|x_3, x_4)p(x_3|x_4)p(x_4) \quad (3.3.1)$$

An equally valid choice is (see fig(3.3))

$$p(x_1, x_2, x_3, x_4) = p(x_3|x_4, x_1, x_2)p(x_4|x_1, x_2)p(x_1|x_2)p(x_2). \quad (3.3.2)$$

In general, two different graphs may represent the same independence assumptions, as we will discuss further in section(3.3.1). If one wishes to make independence assumptions, then the choice of factorisation becomes significant.

The observation that any distribution may be written in the *cascade* form, fig(3.3), gives an algorithm for constructing a BN on variables $x_1, \dots, x_n$: write down the n -variable cascade graph; assign any ordering of the variables to the nodes; you may then delete any of the directed connections. More formally, this corresponds to an ordering of the variables which, without loss of generality, we may write as $x_1, \dots, x_n$. Then, from Bayes’ rule, we have

$$p(x_1, \dots, x_n) = p(x_1|x_2, \dots, x_n)p(x_2, \dots, x_n) \quad (3.3.3)$$

$$= p(x_1|x_2, \dots, x_n)p(x_2|x_3, \dots, x_n)p(x_3, \dots, x_n) \quad (3.3.4)$$

$$= p(x_n) \prod_{i=1}^{n-1} p(x_i|x_{i+1}, \dots, x_n) \quad (3.3.5)$$

The representation of any BN is therefore a *Directed Acyclic Graph (DAG)*.

Every probability distribution can be written as a BN, even though it may correspond to a fully connected ‘cascade’ DAG. The particular role of a BN is that the structure of the DAG corresponds to a set of conditional independence assumptions, namely which parental variables are sufficient to specify each conditional probability table. Note that this does not mean that non-parental variables have no influence. For example, for distribution $p(x_1|x_2)p(x_2|x_3)p(x_3)$ with DAG $x_1 \leftarrow x_2 \leftarrow x_3$, this does not imply $p(x_2|x_1, x_3) = p(x_2|x_3)$. The DAG specifies conditional independence statements of variables on their ancestors – namely which ancestors are ‘causes’ for the variable.

The DAG corresponds to a statement of conditional independencies in the model. To complete the specification of the BN we need to define all elements of the conditional probability tables $p(x_i|\text{pa}(x_i))$. Once the graphical structure is defined, the entries of the conditional probability tables (CPTs) $p(x_i|\text{pa}(x_i))$ can be expressed. For every possible state of the parental variables $\text{pa}(x_i)$, a value for each of the states of x_i needs to be specified (except one, since this is determined by normalisation). For a large number of parents, writing out a table of values is intractable, and the tables are usually parameterised in a low dimensional manner. This will be a central topic of our discussion on the application of BNs in machine learning.

3.3.1 Conditional independence

Whilst a BN corresponds to a set of conditional independence assumptions, it is not always immediately clear from the DAG whether a set of variables is conditionally independent of a set of other variables. For example, in fig(3.4) are x_1 and x_2 independent, given the state of x_4 ? The answer is yes, since we have

$$p(x_1, x_2 | x_4) = \frac{1}{p(x_4)} \sum_{x_3} p(x_1, x_2, x_3, x_4) = \frac{1}{p(x_4)} \sum_{x_3} p(x_1 | x_4) p(x_2 | x_3, x_4) p(x_3) p(x_4) \quad (3.3.6)$$

$$= p(x_1 | x_4) \sum_{x_3} p(x_2 | x_3, x_4) p(x_3) \quad (3.3.7)$$

Now

$$p(x_2 | x_4) = \frac{1}{p(x_4)} \sum_{x_1, x_3} p(x_1, x_2, x_3, x_4) = \frac{1}{p(x_4)} \sum_{x_1, x_3} p(x_1 | x_4) p(x_2 | x_3, x_4) p(x_3) p(x_4) \quad (3.3.8)$$

$$= \sum_{x_3} p(x_2 | x_3, x_4) p(x_3) \quad (3.3.9)$$

Combining the two results above we have

$$p(x_1, x_2 | x_4) = p(x_1 | x_4) p(x_2 | x_4) \quad (3.3.10)$$

so that x_1 and x_2 are indeed independent conditioned on x_4 .

Definition 19 (Conditional Independence).

$$\mathcal{X} \perp\!\!\!\perp \mathcal{Y} | \mathcal{Z} \quad (3.3.11)$$

denotes that the two sets of variables $\mathcal{X}$ and $\mathcal{Y}$ are independent of each other provided we know the state of the set of variables $\mathcal{Z}$. For full conditional independence, $\mathcal{X}$ and $\mathcal{Y}$ must be independent given *all* states of $\mathcal{Z}$. Formally, this means that

$$p(\mathcal{X}, \mathcal{Y} | \mathcal{Z}) = p(\mathcal{X} | \mathcal{Z}) p(\mathcal{Y} | \mathcal{Z}) \quad (3.3.12)$$

for all states of $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$. In case the conditioning set is empty we may also write $\mathcal{X} \perp\!\!\!\perp \mathcal{Y}$ for $\mathcal{X} \perp\!\!\!\perp \mathcal{Y} | \emptyset$, in which case $\mathcal{X}$ is (unconditionally) independent of $\mathcal{Y}$.

If $\mathcal{X}$ and $\mathcal{Y}$ are *not* conditionally independent, they are conditionally dependent. This is written

$$\mathcal{X} \not\perp\!\!\!\perp \mathcal{Y} | \mathcal{Z} \quad (3.3.13)$$

To develop intuition about conditional independence consider the three variable distribution $p(x_1, x_2, x_3)$. We may write this in any of the 6 ways

$$p(x_1, x_2, x_3) = p(x_{i_1} | x_{i_2}, x_{i_3}) p(x_{i_2} | x_{i_3}) p(x_{i_3}) \quad (3.3.14)$$

where (i_1, i_2, i_3) is any of the 6 permutations of $(1, 2, 3)$. Whilst all different DAGs, they represent the same distribution, namely that which makes no conditional independence statements.

To make an independence statement, we need to drop one of the links. This gives rise to the 4 DAGs in fig(3.5). Are any of these graphs equivalent, in the sense that they represent the same distribution?

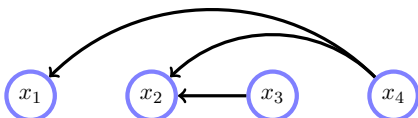


Figure 3.4: $p(x_1, x_2, x_3, x_4) = p(x_1 | x_4) p(x_2 | x_3, x_4) p(x_3) p(x_4)$.

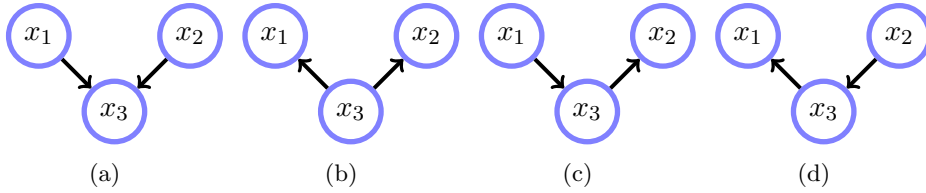


Figure 3.5: By dropping say the connection between variables x_1 and x_2 , we reduce the 6 possible BN graphs amongst three variables to 4. (The 6 fully connected ‘cascade’ graphs correspond to (a) with $x_1 \rightarrow x_2$, (a) with $x_2 \rightarrow x_1$, (b) with $x_1 \rightarrow x_2$, (b) with $x_2 \rightarrow x_1$, (c) with $x_1 \rightarrow x_3$ and (d) with $x_2 \rightarrow x_1$. Any other graphs would be cyclic and therefore not distributions).

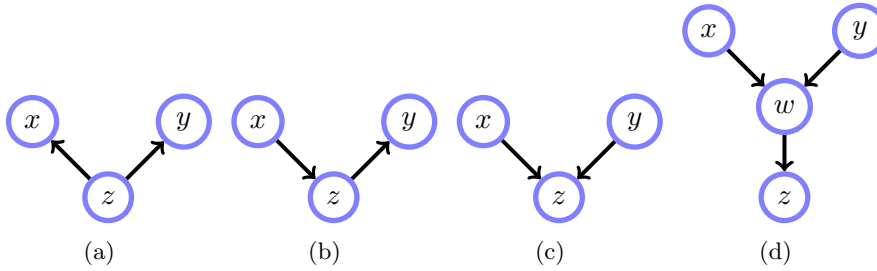


Figure 3.6: In graphs (a) and (b), variable z is not a collider. (c): Variable z is a collider. Graphs (a) and (b) represent conditional independence $x \perp\!\!\!\perp y | z$. In graphs (c) and (d), x and y are ‘graphically’ conditionally dependent given variable z .

Applying Bayes’ rule gives :

$$\underbrace{p(x_2|x_3)p(x_3|x_1)p(x_1)}_{\text{graph}(c)} = p(x_2, x_3)p(x_3, x_1)/p(x_3) = p(x_1|x_3)p(x_2, x_3) \quad (3.3.15)$$

$$= \underbrace{p(x_1|x_3)p(x_3|x_2)p(x_2)}_{\text{graph}(d)} = \underbrace{p(x_1|x_3)p(x_2|x_3)p(x_3)}_{\text{graph}(b)} \quad (3.3.16)$$

so that DAGs (b),(c) and (d) represent the same CI assumptions namely that, given the state of variable x_3 , variables x_1 and x_2 are independent, $x_1 \perp\!\!\!\perp x_2 | x_3$.

However, graph (a) represents something fundamentally different, namely: $p(x_1, x_2) = p(x_1)p(x_2)$. There is no way to transform the distribution $p(x_3|x_1, x_2)p(x_1)p(x_2)$ into any of the others.

Remark 3 (Graphical Dependence). Belief networks are good for encoding conditional independence, but are not well suited to describing dependence. For example, consider the trivial network $p(a, b) = p(b|a)p(a)$ which has the DAG representation $a \rightarrow b$. This may appear to encode that a and b are dependent. However, there are certainly instances when this is not the case. For example, it may be that the conditional is such that $p(b|a) = p(b)$, so that $p(a, b) = p(a)p(b)$. In this case, although generally there may appear to be a ‘graphical’ dependence from the DAG, there can be instances of the distributions for which dependence does not follow. The same holds for Markov networks, section(4.2), in which $p(a, b) = \phi(a, b)$. Whilst this suggests ‘graphical’ dependence between a and b , for $\phi(a, b) = \phi(a)\phi(b)$, the variables are independent.

3.3.2 The impact of collisions

In a general BN, how could we check if $x \perp\!\!\!\perp y | z$? In fig(3.6)(a,b), x and y are independent when conditioned on z . In fig(3.6)(c) they are dependent; in this situation, variable z is called a *collider* – the arrows of its

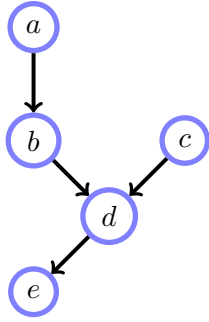


Figure 3.7: The variable d is a collider along the path $a-b-d-c$, but not along the path $a-b-d-e$. Is $a \perp\!\!\!\perp e|b$? a and b are *not* d-connected since there are no colliders on the only path between a and e , and since there is a non-collider b which is in the conditioning set. Hence a and b are d-separated, *i.e.* $a \perp\!\!\!\perp e|b$.

neighbours are pointing towards it. What about fig(3.6)(d)? In (d), when we condition on z , x and y will be ‘graphically’ dependent, since

$$p(x, y|z) = \frac{p(x, y, z)}{p(z)} = \frac{1}{p(z)} \sum_w p(z|w)p(w|x, y)p(x)p(y) \neq p(x|z)p(y|z) \quad (3.3.17)$$

– intuitively, variable w becomes dependent on the value of z , and since x and y are conditionally dependent on w , they are also conditionally dependent on z .

Roughly speaking, if there is a non-collider z which is conditioned on along the path between x and y (as in fig(3.6)(a,b)), then this path does not make x and y dependent. Similarly, if there is a path between x and y which contains a collider, provided that this collider is not in the conditioning set (and neither are any of its descendants) then this path does not make x and y dependent. If there is a path between x and y which contains no colliders and no conditioning variables, then this path ‘d-connects’ x and y .

Note that a collider is defined *relative to a path*. In fig(3.7), the variable d is a collider along the path $a-b-d-c$, but not along the path $a-b-d-e$ (since, relative to this path, the two arrows do not point inwards to d).

Consider the BN: $A \rightarrow B \leftarrow C$. Here A and C are (unconditionally) independent. However, conditioning of B makes them ‘graphically’ dependent. Intuitively, whilst we believe the root causes are independent, given the value of the observation, this tells us something about the state of *both* the causes, coupling them and making them (generally) dependent.

3.3.3 d-Separation

The DAG concepts of d-separation and d-connection are central to determining conditional independence in any BN with structure given by the DAG[284].

Definition 20 (d-connection, d-separation). If G is a directed graph in which $\mathcal{X}$, $\mathcal{Y}$ and $\mathcal{Z}$ are disjoint sets of vertices, then $\mathcal{X}$ and $\mathcal{Y}$ are d-connected by $\mathcal{Z}$ in G if and only if there exists an undirected path U between some vertex in $\mathcal{X}$ and some vertex in $\mathcal{Y}$ such that for every collider C on U , either C or a descendent of C is in $\mathcal{Z}$, and no non-collider on U is in $\mathcal{Z}$.

$\mathcal{X}$ and $\mathcal{Y}$ are d-separated by $\mathcal{Z}$ in G if and only if they are not d-connected by $\mathcal{Z}$ in G .

One may also phrase this as follows. For every variable $x \in \mathcal{X}$ and $y \in \mathcal{Y}$, check every path U between x and y . A path U is said to be *blocked* if there is a node w on U such that either

1. w is a collider and neither w nor any of its descendants is in $\mathcal{Z}$.
2. w is not a collider on U and w is in $\mathcal{Z}$.

If all such paths are blocked then $\mathcal{X}$ and $\mathcal{Y}$ are d-separated by $\mathcal{Z}$.

If the variable sets $\mathcal{X}$ and $\mathcal{Y}$ are d-separated by $\mathcal{Z}$, they are independent conditional on $\mathcal{Z}$ in all probability distributions such a graph can represent.

The Bayes Ball algorithm[241] provides a linear time complexity algorithm which given a set of nodes $\mathcal{X}$ and $\mathcal{Z}$ determines the set of nodes $\mathcal{Y}$ such that $\mathcal{X} \perp\!\!\!\perp \mathcal{Y} | \mathcal{Z}$. $\mathcal{Y}$ is called the set of irrelevant nodes for $\mathcal{X}$ given $\mathcal{Z}$.

3.3.4 d-Connection and dependence

Given a DAG we can imply with certainty that two variables are (conditionally) independent, provided they are d-separated. Can we infer that they are dependent, provided they are d-connected? Consider the following situation

$$p(a, b, c) = p(c|a, b)p(a)p(b) \quad (3.3.18)$$

for which we note that a and b are d-connected by c . For concreteness, we assume c is binary with states 1,2. The question is whether a and b are dependent, conditioned on c , $a \perp\!\!\!\perp b | c$. To answer this, consider

$$p(a, b|c = 1) = \frac{p(c = 1|a, b)p(a)p(b)}{\sum_{a,b} p(c = 1|a, b)p(a)p(b)} \quad (3.3.19)$$

In general, the first term $p(c = 1|a, b)$ does not need to be a factored function of a and b and therefore a and b are conditionally ‘graphically’ dependent. However, we can construct cases where this is not so. For example, let

$$p(c = 1|a, b) = \phi(a)\psi(b), \text{ and } p(c = 2|a, b) = 1 - p(c = 1|a, b) \quad (3.3.20)$$

where $\phi(a)$ and $\psi(b)$ are arbitrary potentials between 0 and 1. Then

$$p(a, b|c = 1) = \frac{1}{Z} \phi(a)p(a)\psi(b)p(b), \quad Z = \sum_a \phi(a)p(a) \sum_b \psi(b)p(b) \quad (3.3.21)$$

which shows that $p(a, b|c = 1)$ is a product of a function in a and function in b , so that a and b are independent, conditioned on $c = 1$.

A second example is given by the distribution

$$p(a, b, c) = p(c|b)p(b|a)p(a) \quad (3.3.22)$$

in which a and c are d-connected by b . The question is, are a and c dependent, $a \perp\!\!\!\perp c | \emptyset$? For simplicity we assume b takes the two states 1,2. Then

$$p(a, c) = p(a) \sum_b p(c|b)p(b|a) = p(a) (p(c|b = 1)p(b = 1|a) + p(c|b = 2)p(b = 2|a)) \quad (3.3.23)$$

$$= p(a)p(b = 1|a) \left[p(c|b = 1) + p(c|b = 2) \left(\frac{1}{p(b = 1|a)} - 1 \right) \right] \quad (3.3.24)$$

For the setting $p(b = 1|a) = \gamma$, for some constant γ for all states of a , then

$$p(a, c) = p(a)\gamma \left(p(c|b = 1) + p(c|b = 2) \left(\frac{1}{\gamma} - 1 \right) \right) \quad (3.3.25)$$

which is a product of a function of a and a function of c . Hence a and c are independent.

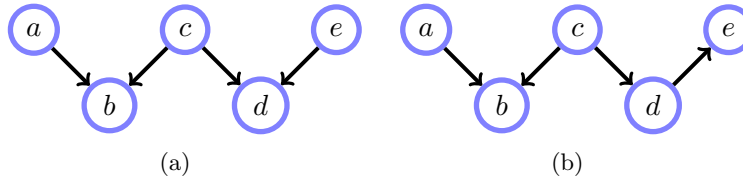


Figure 3.8: Examples for d-separation. (a): b d-separates a from e . The joint variables $\{b, d\}$ d-connect a and e . (b): c and e are (unconditionally) d-connected. b d-connects a and e .

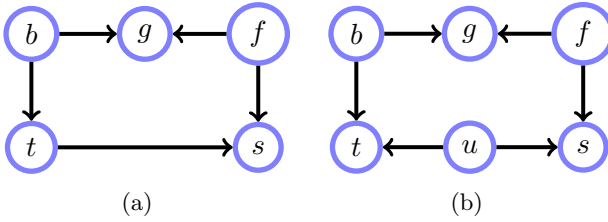


Figure 3.9: (a): t and f are d-connected by g . (b): b and f are d-separated by u .

The moral of the story is that d-separation necessarily implies independence. However, d-connection does not necessarily imply dependence. It might be that there are numerical settings for which variables are independent, even though they are d-connected. For this reason we use the term ‘graphical’ dependence when the graph would suggest that variables are dependent, even though there may be numerical instantiations where dependence does not hold, see definition(21).

Example 12. Consider fig(3.8a). Is $a \perp\!\!\!\perp e \mid b$? If we sum out variable d , then we see that a and e are independent given b , since the variable e will appear as an isolated factor independent of all other variables, hence indeed $a \perp\!\!\!\perp e \mid b$. Whilst b is a collider which is in the conditioning set, we need all colliders on the path to be in the conditioning set (or their descendants) for d-connectedness.

In fig(3.8b), if we sum out variable d , then c and e become intrinsically linked and $p(a, b, c, e)$ will not factorise into a function of a multiplied by a function of e – hence they are dependent.

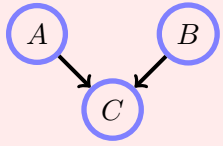
Example 13. Consider the graph in fig(3.9a).

1. Are the variables t and f unconditionally independent, i.e. $t \perp\!\!\!\perp f \mid \emptyset$? Here there are two colliders, namely g and s – however, these are not in the conditioning set (which is empty), and hence they are d-separated and therefore unconditionally independent.
2. What about $t \perp\!\!\!\perp f \mid g$? There is a collider on the path between t and f which is in the conditioning set. Hence t and f are d-connected by g , and therefore t and f are not independent conditioned on g .
3. What about $b \perp\!\!\!\perp f \mid s$? Since there is a collider s in the conditioning set on the path between t and f , then b and f are ‘graphically’ conditionally dependent given s .

Example 14. Is $\{b, f\} \perp\!\!\!\perp u \mid \emptyset$ in fig(3.9b). Since the conditioning set is empty and every path from either b or f to u contains a collider, b, f are unconditionally independent of u .

3.3.5 Markov equivalence in belief networks

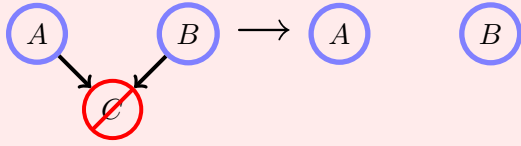
Definition 21 (Some properties of Belief Networks).



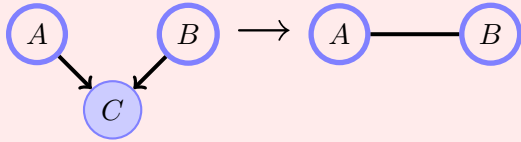
$$p(A, B, C) = p(C|A, B)p(A)p(B) \quad (3.3.26)$$

A and B are (unconditionally) independent : $p(A, B) = p(A)p(B)$.

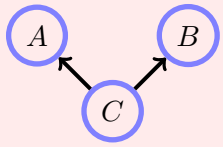
A and B are conditionally dependent on C : $p(A, B|C) \neq p(A|C)p(B|C)$.



Marginalising over C makes A and B independent.



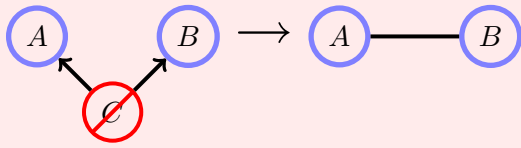
Conditioning on C makes A and B (graphically) dependent.



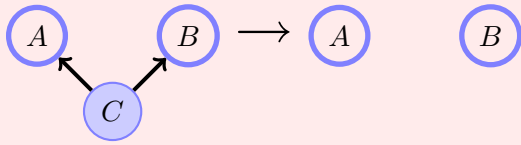
$$p(A, B, C) = p(A|C)p(B|C)p(C) \quad (3.3.27)$$

A and B are (unconditionally) dependent : $p(A, B) \neq p(A)p(B)$.

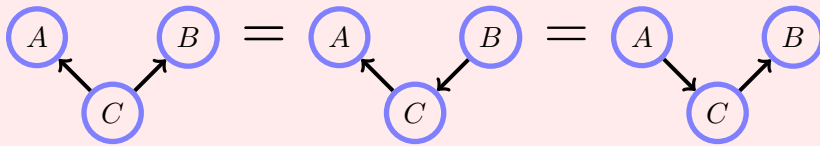
A and B are conditionally independent on C : $p(A, B|C) = p(A|C)p(B|C)$.



Marginalising over C makes A and B (graphically) dependent.



Conditioning on C makes A and B independent.



Definition 22 (Markov Equivalence). Two graphs are Markov equivalent if they both represent the same set of conditional independence statements.

Define the *skeleton* of a graph by removing the directions on the arrows. Define an *immorality* in a DAG as a configuration of three nodes, A, B, C such that C is a child of both A and B , with A and B not directly connected. Two DAGs represent the same set of independence assumptions (they are *Markov equivalent*) if and only if they have the same skeleton and the same set of immoralities [74].

Using this rule we see that in fig(3.5), BNs (b,c,d) have the same skeleton with no immoralities and are therefore equivalent. However BN (a) has an immorality and is therefore not equivalent to DAGS (b,c,d).

3.3.6 Belief networks have limited expressibility

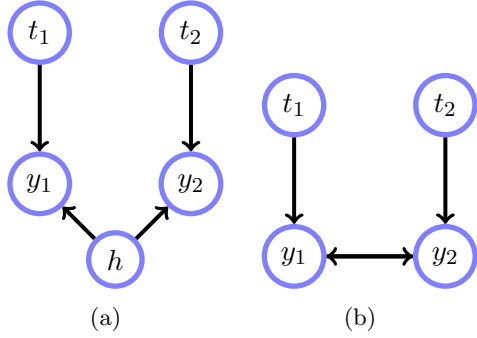


Figure 3.10: **(a)**: Two treatments t_1, t_2 and corresponding outcomes y_1, y_2 . The health of a patient is represented by h . This DAG embodies the conditional independence statements $t_1 \perp\!\!\!\perp t_2, y_2 \mid \emptyset$, $t_2 \perp\!\!\!\perp t_1, y_1 \mid \emptyset$, namely that the treatments have no effect on each other. **(b)**: One could represent the marginalised latent variable using a bi-directional edge.

Consider the DAG in fig(3.10a), (from [232]). This DAG could be used to represent two successive experiments where t_1 and t_2 are two treatments and y_1 and y_2 represent two outcomes of interest; h is the underlying health status of the patient; the first treatment has no effect on the second outcome hence there is no edge from y_1 to y_2 . Now consider the implied independencies in the marginal distribution $p(t_1, t_2, y_1, y_2)$, obtained by marginalising the full distribution over h . There is no DAG containing only the vertices t_1, y_1, t_2, y_2 which represents the independence relations and does not also imply some other independence relation that is not implied by fig(3.10a). Consequently, any DAG on vertices t_1, y_1, t_2, y_2 alone will either fail to represent an independence relation of $p(t_1, t_2, y_1, y_2)$, or will impose some additional independence restriction that is not implied by the DAG. In the above example

$$p(t_1, t_2, y_1, y_2) = p(t_1)p(t_2) \sum_h p(y_1|t_1, h)p(y_2|t_2, h)p(h) \quad (3.3.28)$$

cannot in general be expressed as a product of functions defined on a limited set of the variables. However, it *is* the case that the conditional independence conditions $t_1 \perp\!\!\!\perp t_2, y_2 \mid \emptyset$, $t_2 \perp\!\!\!\perp t_1, y_1 \mid \emptyset$ hold in $p(t_1, t_2, y_1, y_2)$ – they are there, encoded in the form of the conditional probability tables. It is just that we cannot ‘see’ this independence since it is not present in the structure of the marginalised graph (though one can naturally infer this in the larger graph $p(t_1, t_2, y_1, y_2, h)$).

This example demonstrates that BNs cannot express all the conditional independence statements that could be made on that set of variables (the set of conditional independence statements can be increased by considering extra latent variables however). This situation is rather general in the sense that any graphical model has limited expressibility in terms of independence statements[265]. It is worth bearing in mind that Belief Networks may not always be the most appropriate framework to express one’s independence assumptions and intuitions.

A natural consideration is to use a bi-directional arrow when a latent variable is marginalised. For fig(3.10a), one could depict the marginal distribution using a bi-directional edge, fig(3.10b). Similarly a BN with a latent conditioned variable can be represented using an undirected edge. For a discussion of these and related issues, see [232].

3.4 Causality

Causality is a contentious topic and the purpose of this section is make the reader aware of some pitfalls that can occur and which may give rise to erroneous inferences. The reader is referred to [220] and [74] for further details.

The word ‘causal’ is contentious particularly in cases where the model of the data contains no explicit temporal information, so that formally only correlations or dependencies can be inferred. For a distribution $p(a, b)$, we could write this as either (i) $p(a|b)p(b)$ or (ii) $p(b|a)p(a)$. In (i) we might think that b ‘causes’ a , and in (ii) a ‘causes’ b . Clearly, this is not very meaningful since they both represent exactly the same distribution. Formally Belief Networks only make independence statements, not causal ones. Nevertheless, in constructing BNs, it can be helpful to think about dependencies in terms of causation



Figure 3.11: Both (a) and (b) represent the same distribution $p(a, b) = p(a|b)p(b) = p(b|a)p(a)$. (c): The graph represents $p(\text{rain}, \text{grasswet}) = p(\text{grasswet}|\text{rain})p(\text{rain})$. (d): We could equally have written $p(\text{rain}|\text{grasswet})p(\text{grasswet})$, although this appears to be causally non-sensical.

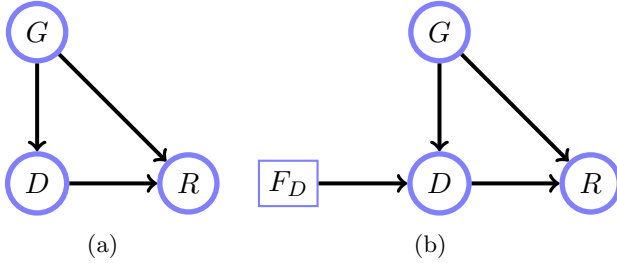


Figure 3.12: (a): A DAG for the relation between Gender (G), Drug (D) and Recovery (R), see table(3.1). (b): Influence diagram. No decision variable is required for G since G has no parents.

since our intuitive understanding is usually framed in how one variable ‘influences’ another. First we discuss a classic conundrum that highlights potential pitfalls that can arise.

3.4.1 Simpson’s paradox

Simpson’s ‘paradox’ is a cautionary tale in causal reasoning in BNs. Consider a medical trial in which patient treatment and outcome are recovered. Two trials were conducted, one with 40 females and one with 40 males. The data is summarised in table(3.1). The question is : Does the drug cause increased recovery? According to the table for males, the answer is no, since more males recovered when they were not given the drug than when they were. Similarly, more females recovered when not given the drug than recovered when given the drug. The conclusion appears that the drug cannot be beneficial since it aids neither subpopulation.

However, ignoring the gender information, and collating both the male and female data into one combined table, we find that more people recovered when given the drug than when not. Hence, even though the drug doesn’t seem to work for either males or females, it does seem to work overall! Should we therefore recommend the drug or not?

Resolution of the paradox

The ‘paradox’ occurs since we are asking a *causal* (or interventional) question. The question we are intuitively asking is, if we give someone the drug, what happens? However, the calculation we performed above was only an observational calculation. The calculation we really want is to first *intervene*, setting

Males	Recovered	Not Recovered	Rec. Rate
Given Drug	18	12	60%
Not Given Drug	7	3	70%

Females	Recovered	Not Recovered	Rec. Rate
Given Drug	2	8	20%
Not Given Drug	9	21	30%

Combined	Recovered	Not Recovered	Rec. Rate
Given Drug	20	20	50%
Not Given Drug	16	24	40%

Table 3.1: Table for Simpson’s Paradox (from [220])

the drug state, and then observe what effect this has on recovery. Pearl[220] describes this as the difference between ‘given that we see’ (observational evidence), versus ‘given that we do’ (interventional evidence).

A model of the Gender, Drug and Recovery data (which makes no conditional independence assumptions) is

$$p(G, D, R) = p(R|G, D)p(D|G)p(G) \quad (3.4.1)$$

An observational calculation concerns computing $p(R|G, D)$ and $p(R|D)$. In a *causal* interpretation, however, if we intervene and give the drug, then the term $p(D|G)$ in equation (3.4.1) should play no role in the experiment (otherwise the distribution models that given the gender we select a drug with probability $p(D|G)$, which is not the case – we decide to give the drug or not, independent of gender). In the causal case we are modelling the *causal experiment*; in this case the term $p(D|G)$ needs to be replaced by a term that reflects the setup of the experiment. In an *atomic intervention* a single variable is set in a particular state⁵. In our atomic causal intervention in setting D , we are dealing with the modified distribution

$$\tilde{p}(G, R|D) = p(R|G, D)p(G) \quad (3.4.2)$$

where the terms on the right hand side of this equation are taken from the original BN of the data. To denote an intervention we use $||$:

$$p(R||G, D) \equiv \tilde{p}(R|G, D) = \frac{p(R|G, D)p(G)}{\sum_R p(R|G, D)p(G)} = p(R|G, D) \quad (3.4.3)$$

(One can also consider here G as being interventional – in this case it doesn’t matter since the fact that the variable G has no parents means that for any distribution conditional on G , the prior factor $p(G)$ will not be present). Using equation (3.4.3), for the males given the drug 60% recover, versus 70% recovery when not given the drug. For the females given the drug 20% recover, versus 30% recovery when not given the drug.

Similarly,

$$p(R||D) \equiv \tilde{p}(R|D) = \frac{\sum_G p(R|G, D)p(G)}{\sum_{R,G} p(R|G, D)p(G)} = \sum_G p(R|G, D)p(G) \quad (3.4.4)$$

Using the above post intervention distribution we have

$$p(\text{recovery}|\text{drug}) = 0.6 \times 0.5 + 0.2 \times 0.5 = 0.4 \quad (3.4.5)$$

and

$$p(\text{recovery}|\text{no drug}) = 0.7 \times 0.5 + 0.3 \times 0.5 = 0.5 \quad (3.4.6)$$

Hence we correctly infer that the drug is overall not helpful, as we intuitively expect, and is consistent with the results from both subpopulations.

Here $p(R||D)$ means that we first choose either a Male or Female patient at random, and then give them the drug, or not depending on the state of D . The point is that we do not randomly decide whether or not to give the drug, hence the absence of the term $p(D|G)$ from the joint distribution. One way to think about such models is to consider how to draw a sample from the joint distribution of the random variables – in most cases this should clarify the role of causality in the experiment.

In contrast to the interventional calculation, the observational calculation makes no conditional independence assumptions. This means that, for example, the term $p(D|G)$ plays a role in the calculation (the reader might wish to verify that the result given in the combined data in table(3.1) is equivalent to inferring with the full distribution equation (3.4.1)).

⁵More general experimental conditions can be modelled by replacing $p(D|G)$ by an intervention distribution $\pi(D|G)$

Definition 23 (Pearl's Do Operator).

In a causal inference, in which the effect of setting variables $X_{c_1}, \dots, X_{c_K}$, $c_k \in \mathcal{C}$, in states $\mathbf{x}_{c_1}, \dots, \mathbf{x}_{c_K}$, is to be inferred, this is equivalent to standard evidential inference in the *post intervention distribution*:

$$p(X|do(X_{c_1} = \mathbf{x}_{c_1}), \dots, do(X_{c_K} = \mathbf{x}_{c_K})) = \frac{p(X_1, \dots, X_n | \mathbf{x}_{c_1}, \dots, \mathbf{x}_{c_K})}{\prod_{i=1}^K p(X_{c_i} | \text{pa}(X_{c_i}))} = \prod_{j \notin \mathcal{C}} p(X_j | \text{pa}(X_j)) \quad (3.4.7)$$

where any parental states of $\text{pa}(X_j)$ of X_j are set in their evidential states. An alternative notation is $p(X | \mathbf{x}_{c_1}, \dots, \mathbf{x}_{c_K})$.

In words, for those variables for which we causally intervene and set in a particular state, the corresponding terms $p(X_{c_i} | \text{pa}(X_{c_i}))$ are removed from the original Belief Network. For variables which are evidential but non-causal, the corresponding factors are not removed from the distribution. The interpretation is that the post intervention distribution corresponds to an experiment in which the causal variables are first set and non-causal variables are subsequently observed.

3.4.2 Influence diagrams and the do-calculus

In making causal inferences we must adjust the model to reflect any causal experimental conditions. In setting any variable into a particular state we need to surgically remove all parental links of that variable. Pearl calls this the *do operator*, and contrasts an observational ('see') inference $p(x|y)$ with a causal ('make' or 'do') inference $p(x|do(y))$.

A useful alternative representation is to append variables X upon which an intervention can possibly be made with a parental decision variable F_X [74]. For example⁶

$$\tilde{p}(D, G, R, F_D) = p(D|F_D, G)p(G)p(R|G, D)p(F_D) \quad (3.4.8)$$

where

$$\begin{aligned} p(D|F_D = \emptyset, G) &\equiv p(D|\text{pa}(D)) \\ p(D|F_D = \mathbf{d}, G) &= 1 \text{ for } D = \mathbf{d} \text{ and } 0 \text{ otherwise} \end{aligned}$$

Hence, if the decision variable F_D is set to the empty state, the variable D is determined by the standard observational term $p(D|\text{pa}(D))$. If the decision variable is set to a state of D , then the variable puts all its probability in that single state of $D = \mathbf{d}$. This has the effect of replacing the conditional probability term a unit factor and any instances of D set to the variable in its interventional state⁷.

A potential advantage of the influence diagram approach over the do-calculus is that deriving conditional independence statements can be made based on standard techniques for the augmented BN. Additionally, for parameter learning, standard techniques apply in which the decision variables are set to the condition under which each data sample was collected (a causal or non-causal sample).

Example 15 (Drivers and Accidents: A causal Belief Network).

⁶Here the Influence Diagram is a distribution over variables including decision variables, in contrast to the application of IDs in chapter(7).

⁷More general cases can be considered in which the variables are placed in a distribution of states [74].

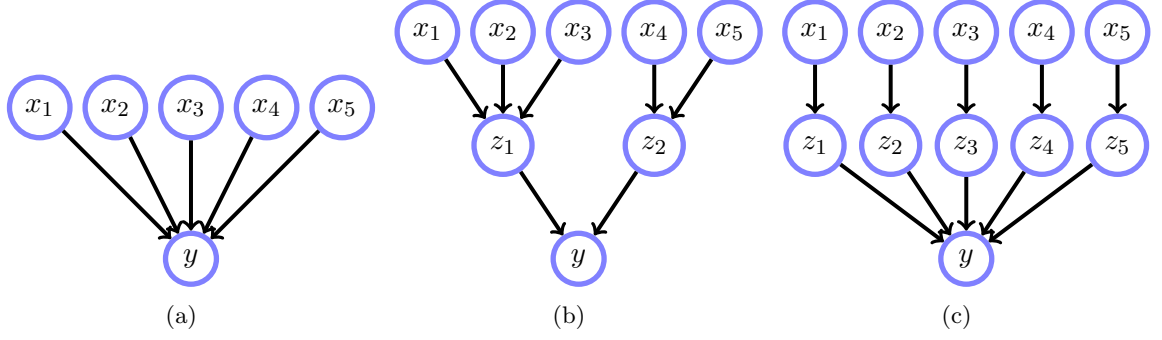
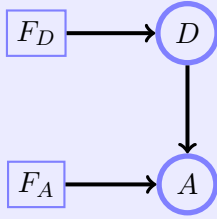


Figure 3.13: **(a)**: If all variables are binary $2^5 = 32$ states are required to specify $p(y|x_1, \dots, x_5)$. **(b)**: Here only 16 states are required. **(c)**: Noisy logic gates.



Consider the following CPT entries $p(D = \text{bad}) = 0.3$, $p(A = \text{tr}|D = \text{bad}) = 0.9$. If we intervene and use a bad driver, what is the probability of an accident?

$$p(A = \text{tr}|D = \text{bad}, F_D = \text{bad}, F_A = \emptyset) = p(A = \text{tr}|D = \text{bad}) = 0.9 \quad (3.4.9)$$

On the other hand, if we intervene and make an accident, what is the probability the driver involved is bad? This is

$$p(D = \text{bad}||A = \text{tr}, F_D = \emptyset, F_A = \text{tr}) = p(D = \text{bad}) = 0.3$$

3.4.3 Learning the direction of arrows

In the absence of data from causal experiments, one should be justifiably sceptical about learning ‘causal’ networks. Nevertheless, one might prefer a certain direction of a link based on assumptions of the ‘simplicity’ of the CPTs. This preference may come from a ‘physical intuition’ that whilst root ‘causes’ may be uncertain, the relationship from cause to effect is clear. In this sense a measure of the complexity of a CPT is required, such as entropy. Such heuristics can be numerically encoded and the ‘directions’ learned in an otherwise Markov equivalent graph.

3.5 Parameterising Belief Networks

Consider a variable y with many parental variables $x_1, \dots, x_n$, fig(3.13a). Formally, the structure of the graph implies nothing about the form of the parameterisation of the table $p(y|x_1, \dots, x_n)$. If each parent x_i has $\dim(x_i)$ states, and there is no constraint on the table, then the table $p(y|x_1, \dots, x_n)$ contains $(\dim(y) - 1) \prod_i \dim(x_i)$ entries. If stored explicitly for each state, this would require potentially huge storage. An alternative is to constrain the table to have a simpler parametric form. For example, one might write a decomposition in which only a limited number of parental interactions are required (this is called *divorcing parents* in [148]). For example, in fig(3.13b), assuming all variables are binary, the number of states requiring specification is $2^3 + 2^2 + 2^2 = 16$, compared to the $2^5 = 32$ states in the unconstrained case. The distribution

$$p(y|x_1, \dots, x_5) = \sum_{z_1, z_2} p(y|z_1, z_2)p(z_1|x_1, x_2, x_3)p(z_2|x_4, x_5) \quad (3.5.1)$$

can be stored using only 16 independent parameters.

Logic gates

Another technique to constrain CPTs uses simple classes of conditional tables. For example, in fig(3.13c), one could use a logical OR gate on binary z_i , say

$$p(y|z_1, \dots, z_5) = \begin{cases} 1 & \text{if at least one of the } z_i \text{ is in state 1} \\ 0 & \text{otherwise} \end{cases} \quad (3.5.2)$$

We can then make a CPT $p(y|x_1, \dots, x_5)$ by including the additional terms $p(z_i = 1|x_i)$. When each x_i is binary there are in total only $2 + 2 + 2 + 2 + 2 = 10$ quantities required for specifying $p(y|x)$. In this case, fig(3.13c) can be used to represent any *noisy logic gate*, such as the *noisy OR* or *noisy AND*, where the number of parameters required to specify the noisy gate is linear in the number of parents x .

The noisy-OR is particularly common in disease-symptom networks in which many diseases x can give rise to the same symptom y — provided that at least one of the diseases is present, the probability that the symptom will be present is high.

3.6 Further Reading

An introduction to Bayesian Networks and graphical models in expert systems is to be found in [258], which also discusses general inference techniques which will be discussed during later chapters.

3.7 Code

3.7.1 Naive inference demo

`demoBurglar.m`: Was it the Burglar demo
`demoChestClinic.m`: Naive Inference on Chest Clinic

3.7.2 Conditional independence demo

The following demo determines whether $\mathcal{X} \perp\!\!\!\perp \mathcal{Y} | \mathcal{Z}$ for the Chest Clinic network, and checks the result numerically⁸. The independence test is based on the Markov method of section(4.2.4). This is preferred over the d-separation method since it is arguably simpler to code and also more general in that it deals also with conditional independence in Markov Networks as well as Belief Networks.

Running the demo code below, it may happen that the numerical dependence is very low – that is

$$p(\mathcal{X}, \mathcal{Y} | \mathcal{Z}) \approx p(\mathcal{X} | \mathcal{Z})p(\mathcal{Y} | \mathcal{Z}) \quad (3.7.1)$$

even though $\mathcal{X} \not\perp\!\!\!\perp \mathcal{Y} | \mathcal{Z}$. This highlights the difference between ‘structural’ and ‘numerical’ independence.

`condindepPot.m`: Numerical measure of conditional independence
`demoCondindep.m`: Demo of conditional independence (using Markov method)

3.7.3 Utility routines

`dag.m`: Find the DAG structure for a Belief Network

3.8 Exercises

Exercise 20 (Party Animal). *The party animal problem corresponds to the network in fig(3.14). The boss is angry and the worker has a headache – what is the probability the worker has been to a party? To*

⁸The code for (structural) conditional independence is given in chapter(4).

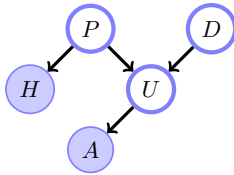
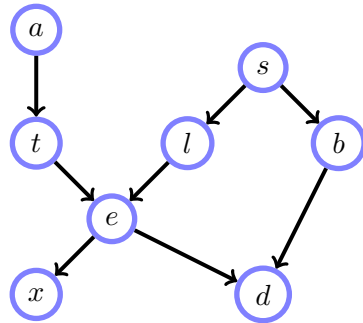


Figure 3.14: Party animal. Here all variables are binary. When set to 1 the statements are true: P = Been to Party, H = Got a Headache, D = Demotivated at work, U = Underperform at work, A = Boss Angry. Shaded variables are observed in the true state.



x = Positive X-ray
 d = Dyspnea (Shortness of breath)
 e = Either Tuberculosis or Lung Cancer
 t = Tuberculosis
 l = Lung Cancer
 b = Bronchitis
 a = Visited Asia
 s = Smoker

Figure 3.15: Belief network structure for the Chest Clinic example.

complete the specifications, the probabilities are given as follows:

$$\begin{aligned} p(U = \text{tr} | P = \text{tr}, D = \text{tr}) &= 0.999 & p(U = \text{tr} | P = \text{fa}, D = \text{tr}) &= 0.9 \\ p(U = \text{tr} | P = \text{tr}, D = \text{fa}) &= 0.9 & p(U = \text{tr} | P = \text{fa}, D = \text{fa}) &= 0.01 \end{aligned}$$

Exercise 21. Consider the distribution $p(a, b, c) = p(c|a, b)p(a)p(b)$. (i) Is $a \perp\!\!\!\perp b | \emptyset$? (ii) Is $a \perp\!\!\!\perp b | c$?

Exercise 22. The Chest Clinic network [170] concerns the diagnosis of lung disease (tuberculosis, lung cancer, or both, or neither). In this model a visit to Asia is assumed to increase the probability of tuberculosis. State if the following conditional independence relationships are true or false

1. tuberculosis $\perp\!\!\!\perp$ smoking | shortness of breath,
2. lung cancer $\perp\!\!\!\perp$ bronchitis | smoking,
3. visit to Asia $\perp\!\!\!\perp$ smoking | lung cancer
4. visit to Asia $\perp\!\!\!\perp$ smoking | lung cancer, shortness of breath.

Exercise 23 ([128]). Consider the network in fig(3.16), which concerns the probability of a car starting.

$$\begin{aligned} p(b = \text{bad}) &= 0.02 & p(f = \text{empty}) &= 0.05 \\ p(g = \text{empty} | b = \text{good}, f = \text{not empty}) &= 0.04 & p(g = \text{empty} | b = \text{good}, f = \text{empty}) &= 0.97 \\ p(g = \text{empty} | b = \text{bad}, f = \text{not empty}) &= 0.1 & p(g = \text{empty} | b = \text{bad}, f = \text{empty}) &= 0.99 \\ p(t = \text{fa} | b = \text{good}) &= 0.03 & p(t = \text{fa} | b = \text{bad}) &= 0.98 \\ p(s = \text{fa} | t = \text{tr}, f = \text{not empty}) &= 0.01 & p(s = \text{fa} | t = \text{tr}, f = \text{empty}) &= 0.92 \\ p(s = \text{fa} | t = \text{fa}, f = \text{not empty}) &= 1.0 & p(s = \text{fa} | t = \text{fa}, f = \text{empty}) &= 0.99 \end{aligned}$$

Calculate $P(f = \text{empty} | s = \text{no})$, the probability of the fuel tank being empty conditioned on the observation that the car does not start.

Exercise 24. Consider the Chest Clinic Bayesian Network in fig(3.15) [170]. Calculate by hand the values for $p(D)$, $p(D | S = \text{tr})$, $p(D | S = \text{fa})$. The table values are:

$$\begin{aligned} p(a = \text{tr}) &= 0.01 & p(s = \text{tr}) &= 0.5 \\ p(t = \text{tr} | a = \text{tr}) &= 0.05 & p(t = \text{tr} | a = \text{fa}) &= 0.01 \\ p(l = \text{tr} | s = \text{tr}) &= 0.1 & p(l = \text{tr} | s = \text{fa}) &= 0.01 \\ p(b = \text{tr} | s = \text{tr}) &= 0.6 & p(b = \text{tr} | s = \text{fa}) &= 0.3 \\ p(x = \text{tr} | e = \text{tr}) &= 0.98 & p(x = \text{tr} | e = \text{fa}) &= 0.05 \\ p(d = \text{tr} | e = \text{tr}, b = \text{tr}) &= 0.9 & p(d = \text{tr} | e = \text{tr}, b = \text{fa}) &= 0.7 \\ p(d = \text{tr} | e = \text{fa}, b = \text{tr}) &= 0.8 & p(d = \text{tr} | e = \text{fa}, b = \text{fa}) &= 0.1 \end{aligned}$$

$p(e = \text{tr} | t, l) = 0$ only if both t and l are fa , 1 otherwise.

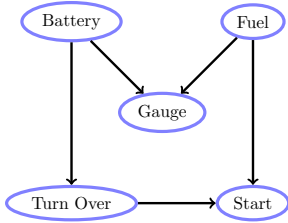


Figure 3.16: Belief Network of car not starting[128], see exercise(23).

Exercise 25. If we interpret the Chest Clinic network exercise(24) causally, how can we help a doctor answer the question ‘If I could cure my patients of Bronchitis, how would this affect my patients’s chance of being short of breath?’. How does this compare with $p(d = \text{tr} | b = \text{fa})$ in a non-causal interpretation, and what does this mean?

Exercise 26. There is a synergistic relationship between Asbestos (A) exposure, Smoking (S) and Cancer (C). A model describing this relationship is given by

$$p(A, S, C) = p(C|A, S)p(A)p(S) \quad (3.8.1)$$

1. Is $A \perp\!\!\!\perp S | \emptyset$?
2. Is $A \perp\!\!\!\perp S | C$?
3. How could you adjust the model to account for the fact that people who work in the building industry have a higher likelihood to also be smokers and also a higher likelihood to asbestos exposure?

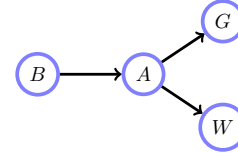
Exercise 27. Consider the three variable distribution

$$p(a, b, c) = p(a|b)p(b|c)p(c) \quad (3.8.2)$$

where all variables are binary. How many parameters are needed to specify distributions of this form?

Exercise 28.

Consider the Belief Network on the right which represents Mr Holmes’ burglary worries as given in fig(3.2a) : (B)urglar, (A)larm, (W)atson, Mrs (G)ibbon).



All variables take the two states $\{\text{tr}, \text{fa}\}$. The table entries are

$$\begin{array}{llll}
 p(B = \text{tr}) & = 0.01 & & \\
 p(A = \text{tr} | B = \text{tr}) & = 0.99 & p(A = \text{tr} | B = \text{fa}) & = 0.05 \\
 p(W = \text{tr} | A = \text{tr}) & = 0.9 & p(W = \text{tr} | A = \text{fa}) & = 0.5 \\
 p(G = \text{tr} | A = \text{tr}) & = 0.7 & p(G = \text{tr} | A = \text{fa}) & = 0.2
 \end{array} \quad (3.8.3)$$

1. Compute ‘by hand’ (i.e. show your working) :

- (a) $p(B = \text{tr} | W = \text{tr})$
- (b) $p(B = \text{tr} | W = \text{tr}, G = \text{fa})$

2. Consider the same situation as above, except that now the evidence is uncertain. Mrs Gibbon thinks that the state is $G = \text{fa}$ with probability 0.9. Similarly, Dr Watson believes in the state $W = \text{fa}$ with value 0.7. Compute ‘by hand’ the posteriors under these uncertain (soft) evidences:

- (a) $p(B = \text{tr} | \tilde{W})$
- (b) $p(B = \text{tr} | \tilde{W}, \tilde{G})$

Exercise 29. A doctor gives a patient a (D)rug (drug or no drug) dependent on their (A)ge (old or young) and (G)ender (male or female). Whether or not the patient (R)ecovers (recovers or doesn’t recover) depends on all D, A, G . In addition $A \perp\!\!\!\perp G | \emptyset$.

1. Write down the Belief Network for the above situation.

2. Explain how to compute $p(\text{recover}|\text{drug})$.
3. Explain how to compute $p(\text{recover}|\text{do}(\text{drug}), \text{young})$.

Exercise 30. Implement the Wet Grass scenario numerically using the BRMLTOOLBOX.

Exercise 31 (LA Burglar). Consider the Burglar scenario, example(10). We now wish to model the fact that in Los Angeles the probability of being burgled increases if there is an earthquake. Explain how to include this effect in the model.

Exercise 32. Given two Belief Networks represented as DAGs with associated adjacency matrices $\mathbf{A}$ and $\mathbf{B}$, write a MATLAB function `MarkovEquiv(A,B).m` that returns 1 if $\mathbf{A}$ and $\mathbf{B}$ are Markov equivalent, and zero otherwise.

Exercise 33. The adjacency matrices of two Belief Networks are given below (see `ABmatrices.mat`). State if they are Markov equivalent.

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad (3.8.4)$$

Exercise 34. There are three computers indexed by $i \in \{1, 2, 3\}$. Computer i can send a message in one timestep to computer j if $C_{ij} = 1$, otherwise $C_{ij} = 0$. There is a fault in the network and the task is to find out some information about the communication matrix $\mathbf{C}$ ($\mathbf{C}$ is not necessarily symmetric). To do this, Thomas, the engineer, will run some tests that reveal whether or not computer i can send a message to computer j in t timesteps, $t \in \{1, 2\}$. This is expressed as $C_{ij}(t)$, with $C_{ij}(1) \equiv C_{ij}$. For example, he might know that $C_{13}(2) = 1$, meaning that according to his test, a message sent from computer 1 will arrive at computer 3 in at most 2 timesteps. Note that this message could go via different routes – it might go directly from 1 to 3 in one timestep, or indirectly from 1 to 2 and then from 2 to 3, or both. You may assume $C_{ii} = 1$. A priori Thomas thinks there is a 10% probability that $C_{ij} = 1$. Given the test information $\mathcal{C} = \{C_{12}(2) = 1, C_{23}(2) = 0\}$, compute the a posteriori probability vector

$$[p(C_{12} = 1|\mathcal{C}), p(C_{13} = 1|\mathcal{C}), p(C_{23} = 1|\mathcal{C}), p(C_{32} = 1|\mathcal{C}), p(C_{21} = 1|\mathcal{C}), p(C_{31} = 1|\mathcal{C})] \quad (3.8.5)$$

Exercise 35. A Belief Network models the relation between the variables *oil*, *inf*, *eh*, *bp*, *rt* which stand for the price of oil, inflation rate, economy health, British Petroleum Stock price, retailer stock price. Each variable takes the states *low*, *high*, except for *bp* which has states *low*, *high*, *normal*. The Belief Network model for these variables has tables

$p(\text{eh}=\text{low})=0.2$	
$p(\text{bp}=\text{low} \text{oil}=\text{low})=0.9$	$p(\text{bp}=\text{normal} \text{oil}=\text{low})=0.1$
$p(\text{bp}=\text{low} \text{oil}=\text{high})=0.1$	$p(\text{bp}=\text{normal} \text{oil}=\text{high})=0.4$
$p(\text{oil}=\text{low} \text{eh}=\text{low})=0.9$	$p(\text{oil}=\text{low} \text{eh}=\text{high})=0.05$
$p(\text{rt}=\text{low} \text{inf}=\text{low}, \text{eh}=\text{low})=0.9$	$p(\text{rt}=\text{low} \text{inf}=\text{low}, \text{eh}=\text{high})=0.1$
$p(\text{rt}=\text{low} \text{inf}=\text{high}, \text{eh}=\text{low})=0.1$	$p(\text{rt}=\text{low} \text{inf}=\text{high}, \text{eh}=\text{high})=0.01$
$p(\text{inf}=\text{low} \text{oil}=\text{low}, \text{eh}=\text{low})=0.9$	$p(\text{inf}=\text{low} \text{oil}=\text{low}, \text{eh}=\text{high})=0.1$
$p(\text{inf}=\text{low} \text{oil}=\text{high}, \text{eh}=\text{low})=0.1$	$p(\text{inf}=\text{low} \text{oil}=\text{high}, \text{eh}=\text{high})=0.01$

1. Draw a Belief Network for this distribution.
2. Given that BP stock price is normal and the retailer stock price is high, what is the probability that inflation is high?

Exercise 36. There are a set of C potentials with potential c defined on a subset of variables $\mathcal{X}_c$. If $\mathcal{X}_c \subseteq \mathcal{X}_d$ then can merge (multiply) potentials c and d since c is contained within d . With reference to suitable graph structures, describe an efficient algorithm to merge a set of potentials so that for the new set of potentials no potential is contained within the other.

